



COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

Computer Networks and the Internet

Protocol Layers

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Unit – 1 Computer Networks and the Internet

1.1 Introduction to Computer Networks

1.2 What is the Internet?

- A nuts-and-bolts and Services description, Protocol

1.3 Network edge

- End systems, Access networks, Physical media

1.4 Network core

- Packet switching, Circuit switching, Network structure

1.5 Delay, Loss & Throughput in networks

1.6 Protocol layers, Service models

- OSI model and TCP/IP protocol suite

Networks are complex,
with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

is there any hope of
organizing structure of
network?

.... or at least our *discussion*
of networks?

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Example: Organization of Air Travel



ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

airline travel: a series of steps, involving many services

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Layering of Airline functionality



layers: each layer implements a service

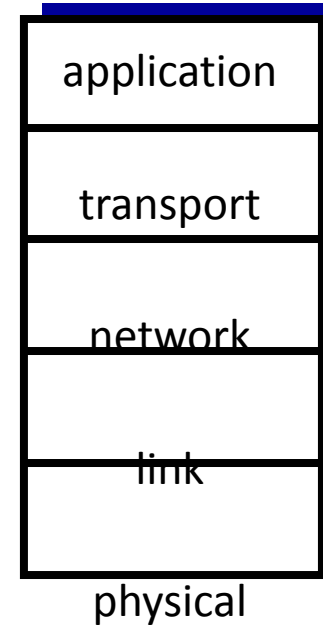
- via its own internal-layer actions
- relying on services provided by layer below

Q: describe in words
the service provided
in each layer above

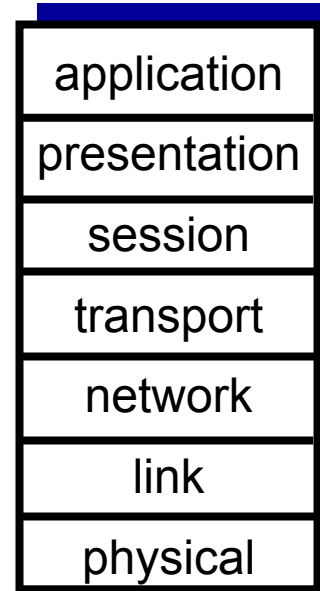
dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces
 - layered *reference model* for discussion
- modularization eases maintenance, updating of system
 - change in layer's service *implementation*: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
- layering considered harmful?
- layering in other complex systems?

- **application:** supporting network applications (access to network resources)
 - IMAP, SMTP, HTTP
- **transport:** process-process data transfer (segmentation & reassembly, sockets, connection, flow and error control)
 - TCP, UDP
- **network:** routing of datagrams from source to destination (addressing, routing)
 - IP, routing protocols
- **link:** data transfer between neighboring network elements (framing, addressing, flow & error control)
 - Ethernet, 802.11 (WiFi), PPP
- **physical:** bits “on the wire”



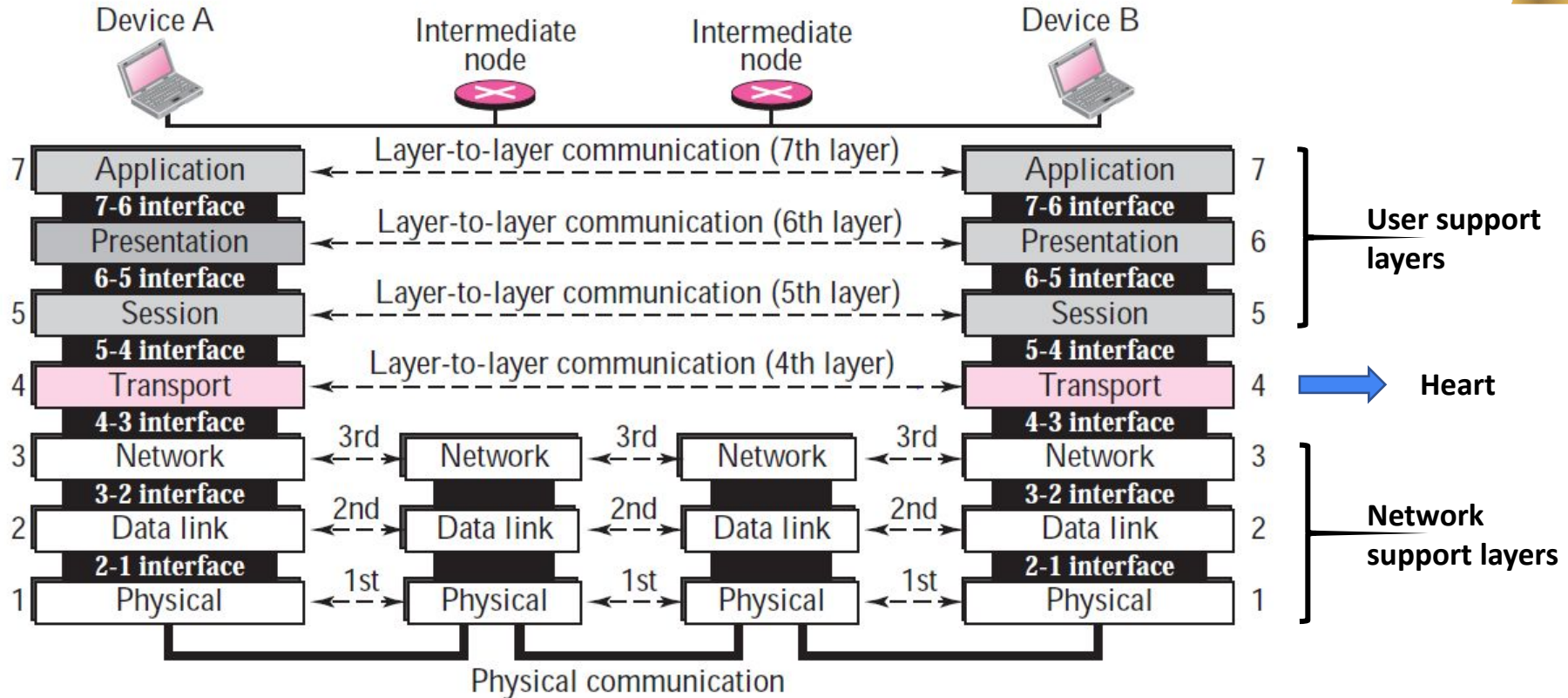
- ***presentation***: allow applications to interpret meaning of data, (e.g., encryption, compression, machine-specific conventions)
- ***session***: synchronization, checkpointing, recovery of data exchange
- Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application
 - needed?



Open Systems Interconnection (OSI) model – introduced in late 1970s by ISO.

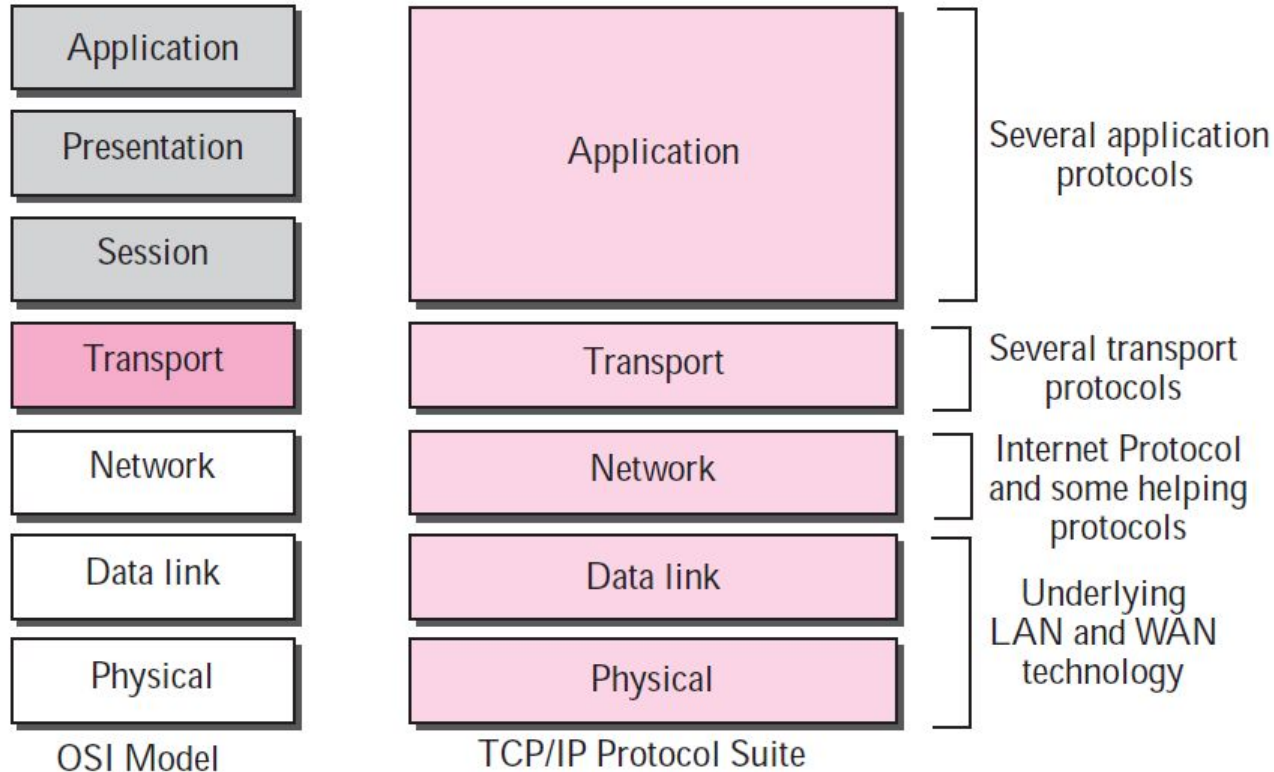
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OSI reference model (more)



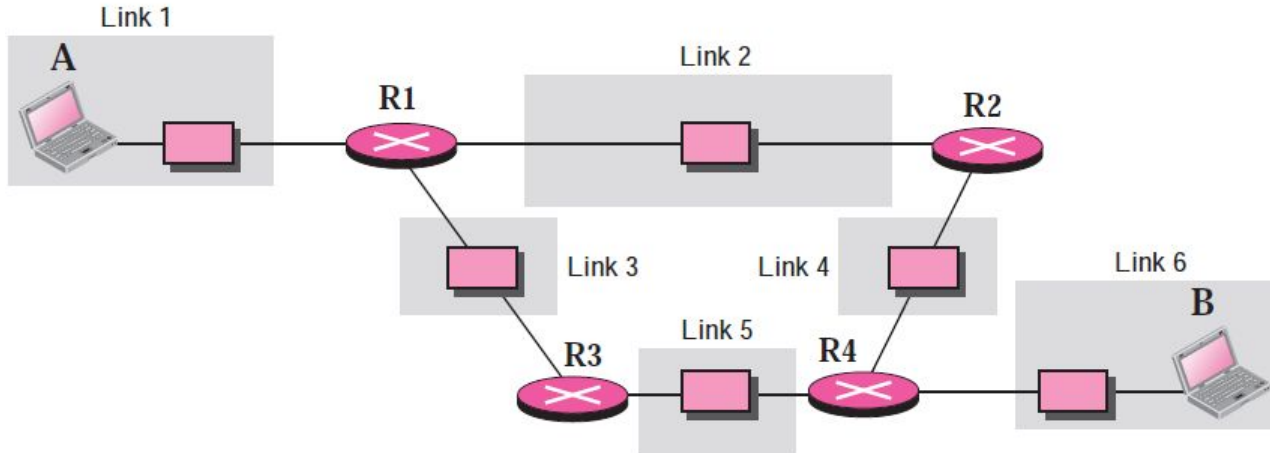
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TCP/IP vs OSI reference model



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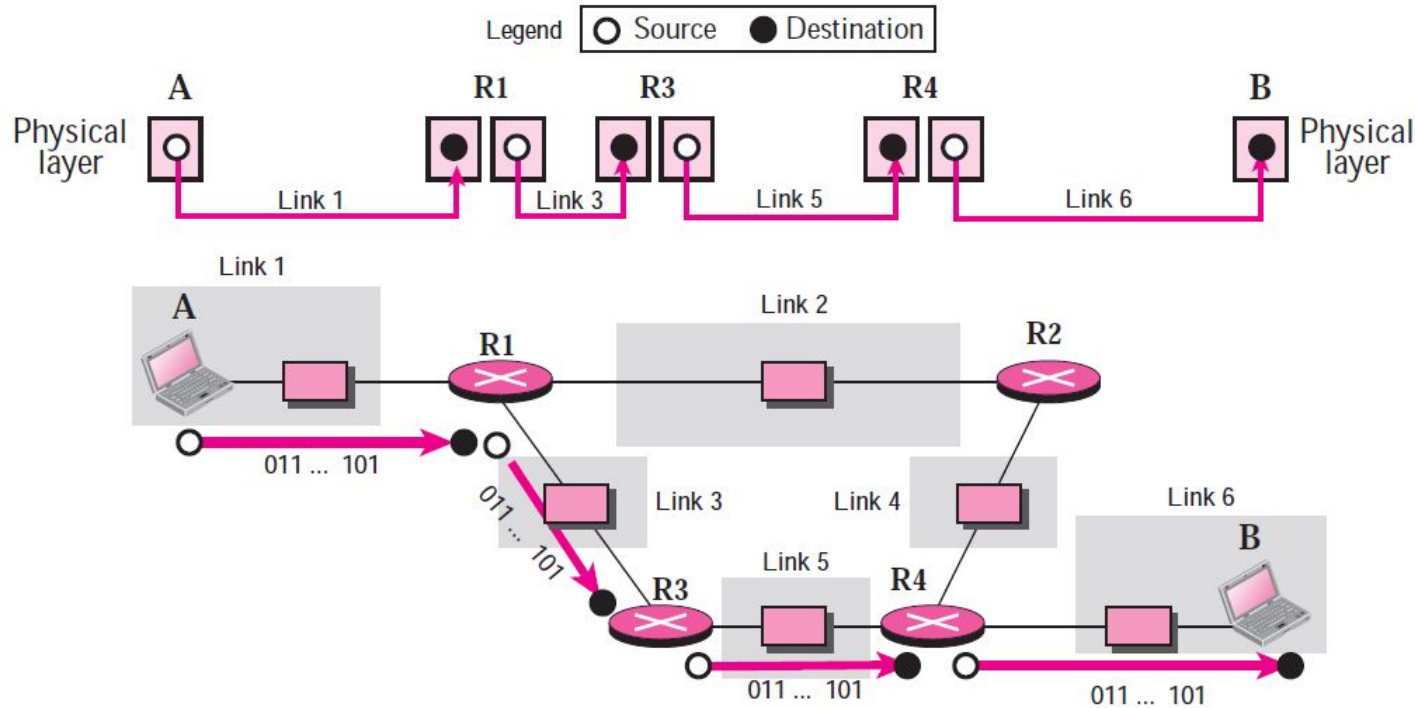
Layers in the TCP/IP Protocol Suite (more)



A private internet

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Layers in the TCP/IP Protocol Suite (more)

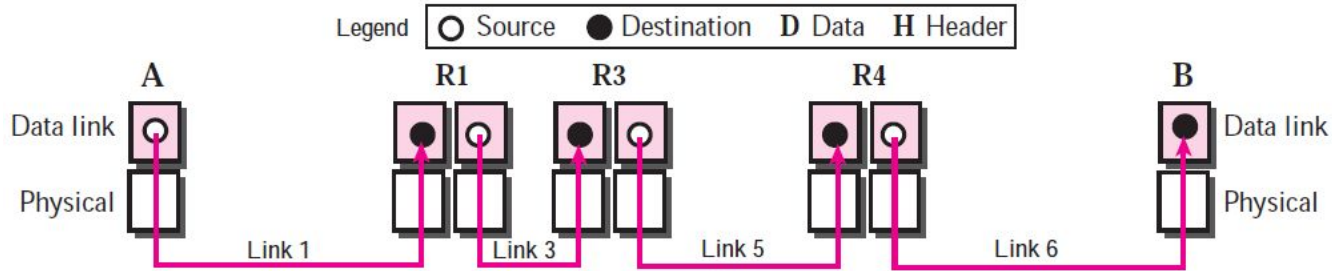


Communication at the physical layer

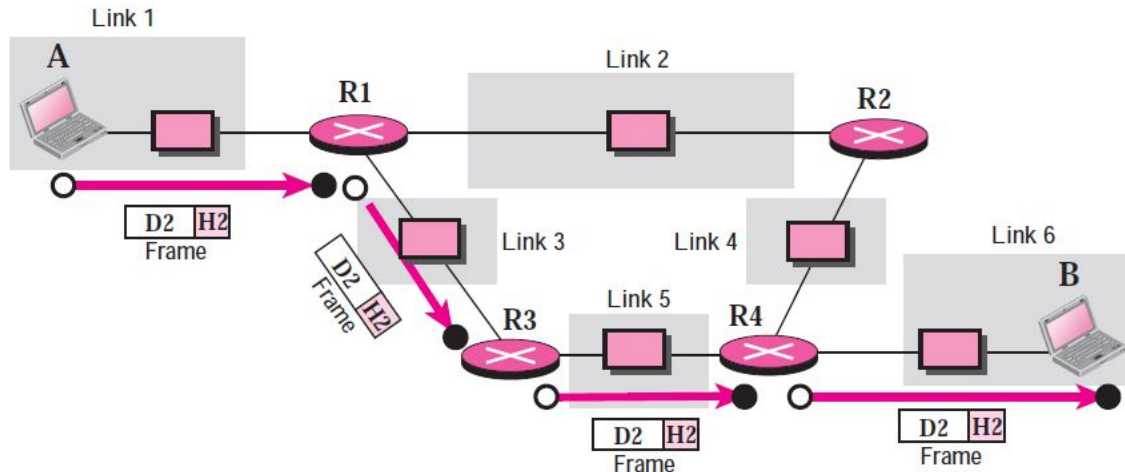
Unit of Communication
– **bit**

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Layers in the TCP/IP Protocol Suite (more)



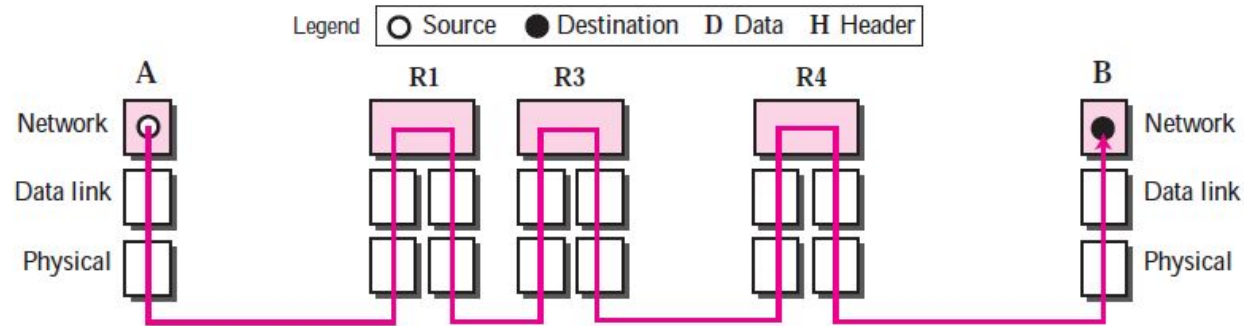
Communication at the data link layer



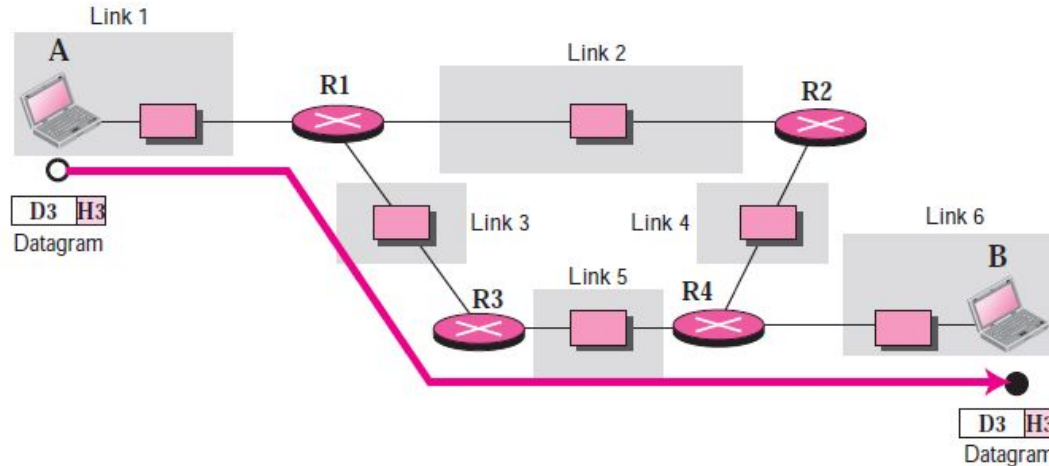
Unit of Communication –
frame

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Layers in the TCP/IP Protocol Suite (more)



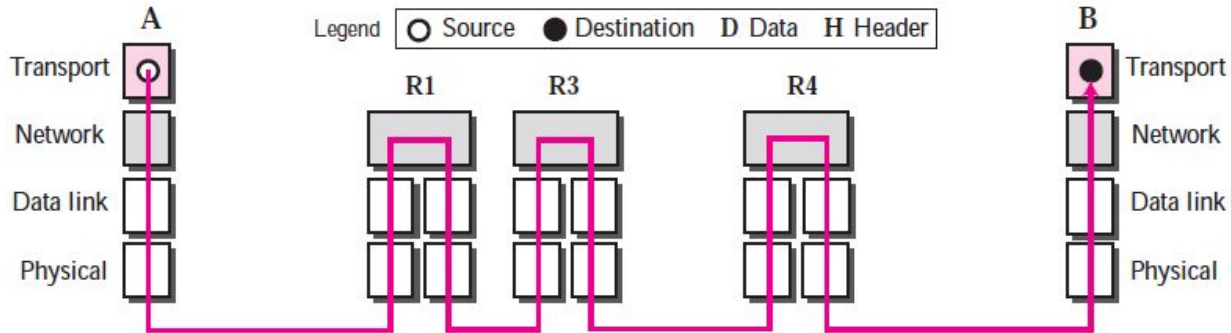
Communication at the network layer



**Unit of Communication –
datagram**

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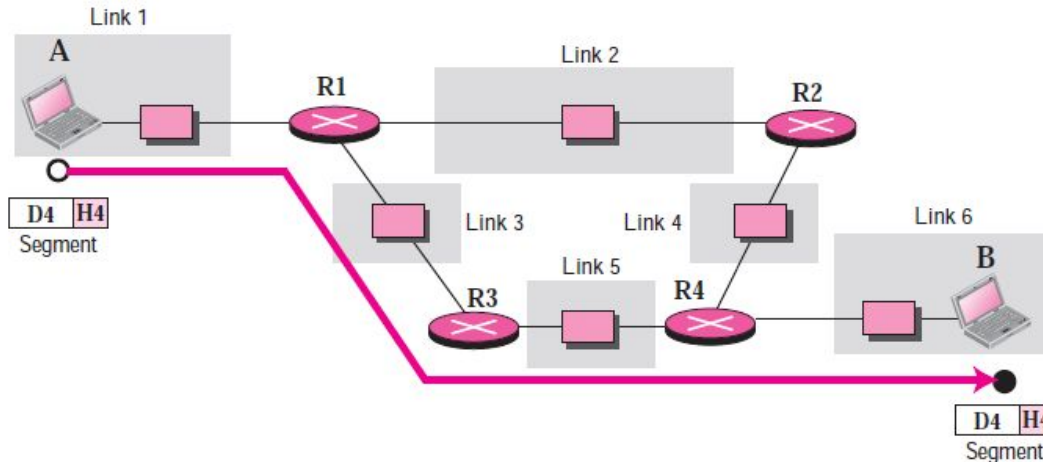
Layers in the TCP/IP Protocol Suite (more)



Communication at the transport layer

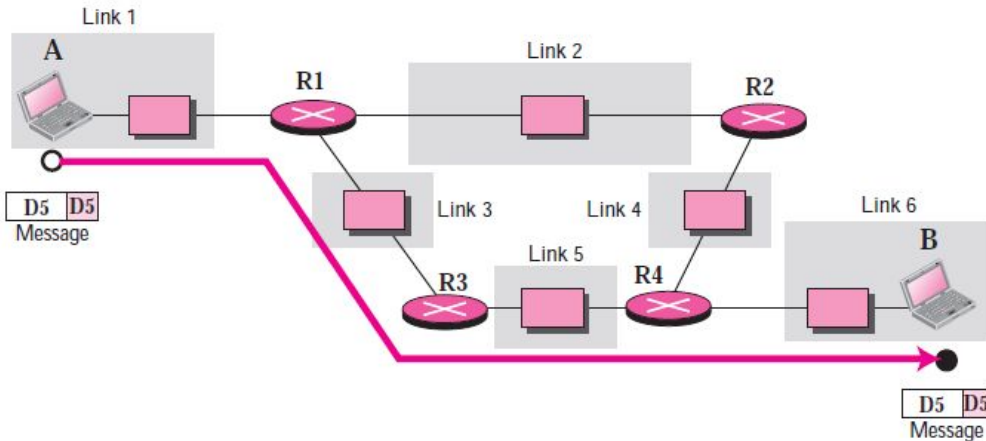
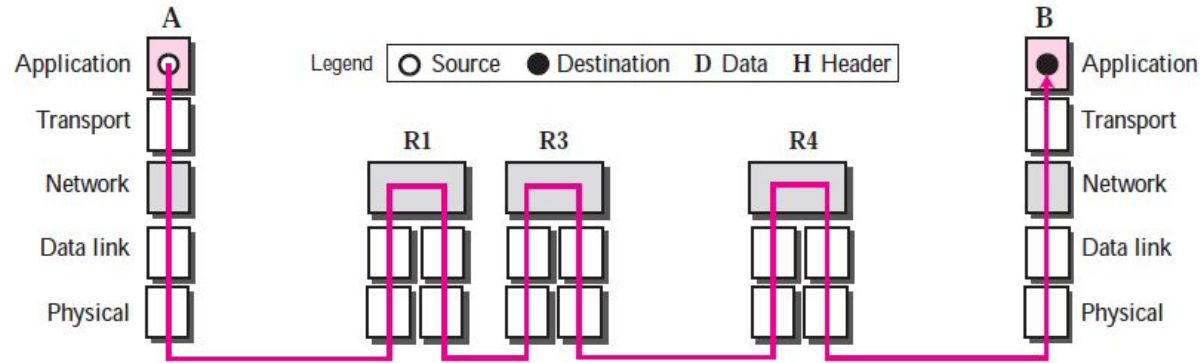
Unit of Communication

– **segment/packet**



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Layers in the TCP/IP Protocol Suite (more)



Communication at the application layer

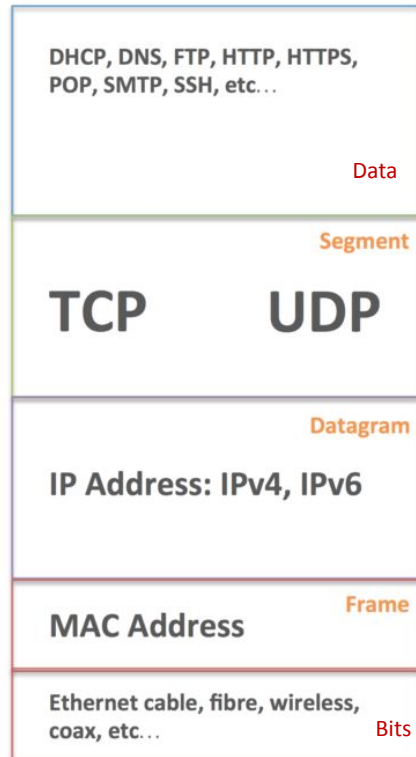
Unit of Communication

– **message**

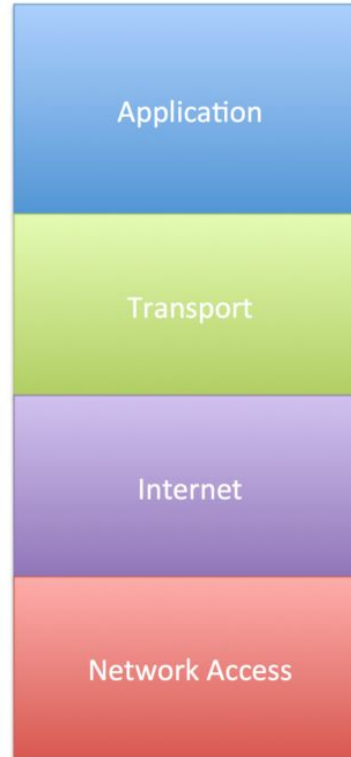
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OSI vs TCP/IP Protocol Suite

The OSI Model

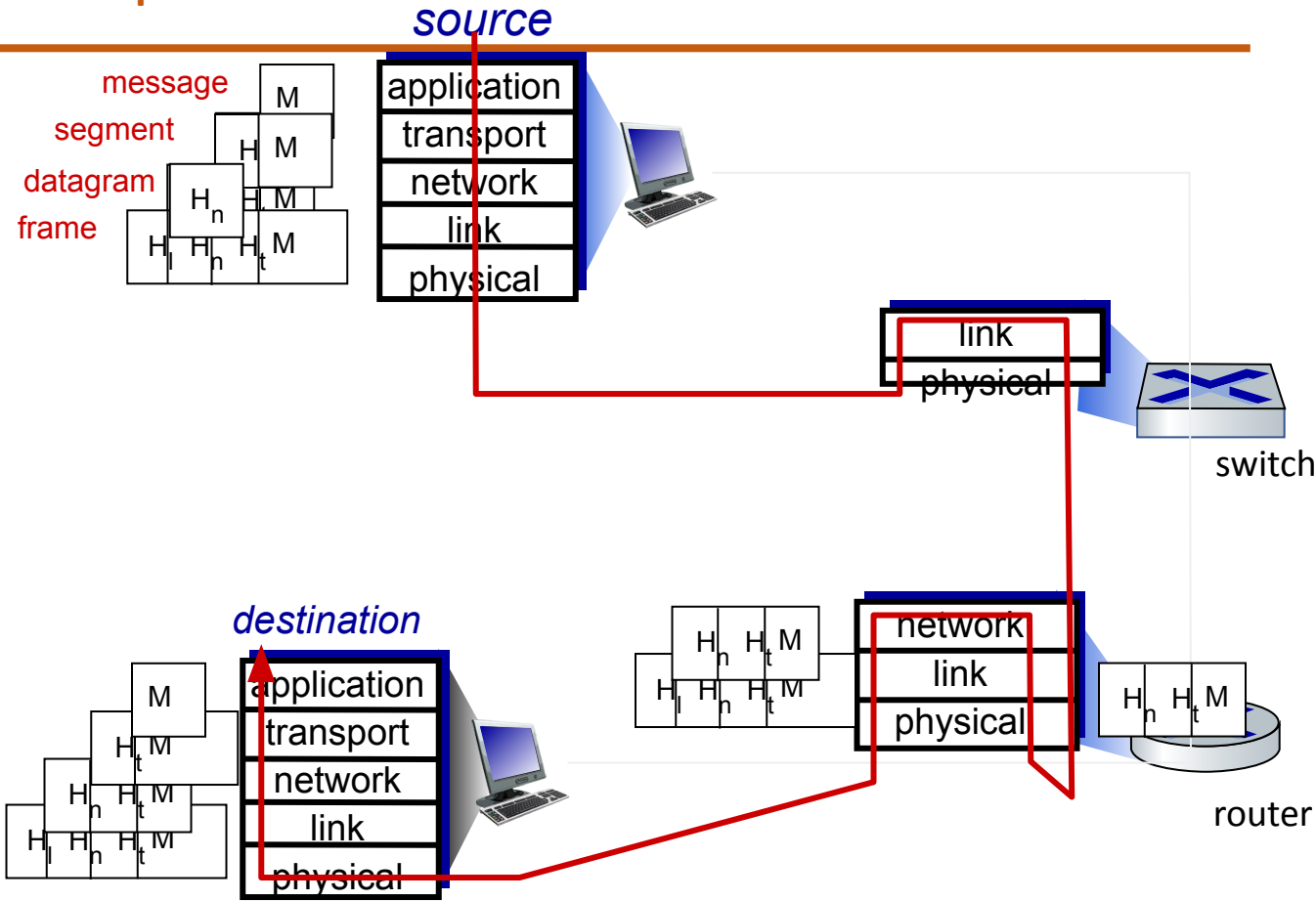


The TCP/IP Model



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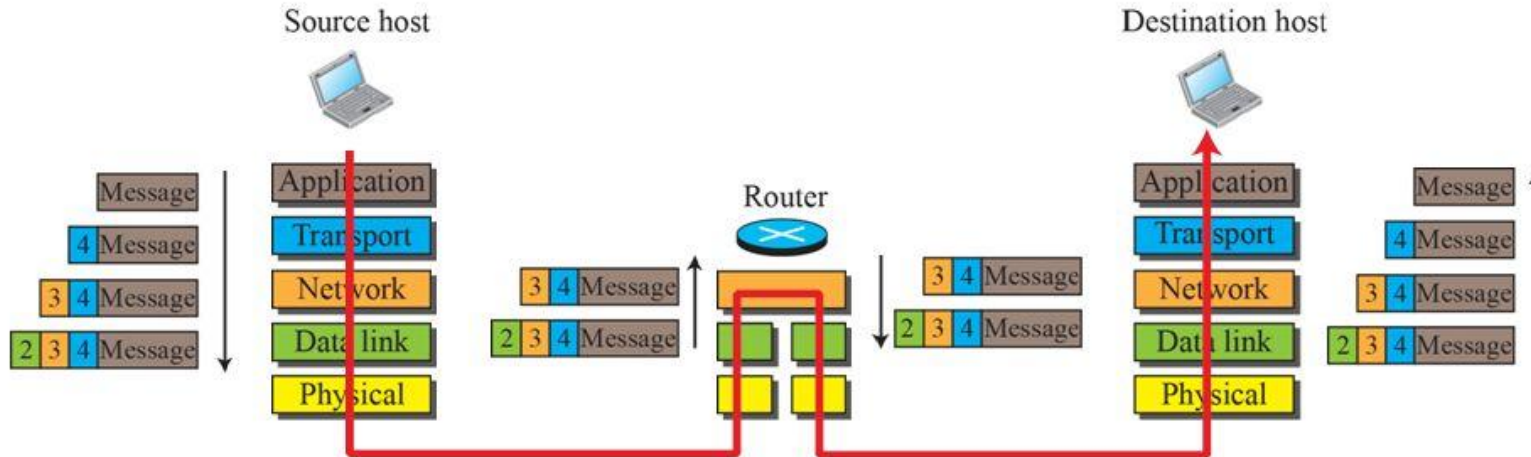
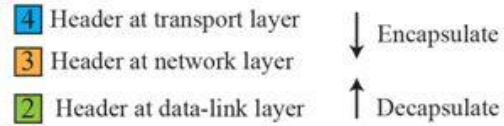
Encapsulation – Data Communication in Protocol Stack



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Encapsulation & Decapsulation

Legend





THANK YOU

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